THADOMAL SHAHANI TSEC ENGINEERING COLLEGE	**THADOMAL SHAHANI ENGINEERING COLLEGE** **DEPARTMENT OF** **Artificial Intelligence & Data Science**

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Experiment Number : 4

Experiment Title : Demonstration of Modem Hardware and Signal Flow

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Experiment No. 04: Demonstration of Modem Hardware and Signal Flow

Aim

To identify the hardware interfaces of a Modem and demonstrate the flow between a PC and an ISP Server using Cisco Packet Tracer.

Theory

1. What is a Modem?

A **Modem** (short for **Modulator-Demodulator**) is a networking device that enables communication between a computer and the internet. It converts digital data from a computer into analog signals that can travel through telephone lines, cable systems, or fiber networks, and converts incoming analog signals back into digital form.

In simple terms, a modem allows your device to connect to the internet by translating signals into a form that can be transmitted over communication channels.

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2. Working Principle of a Modem

A modem works using two main processes:

Modulation

When a computer sends data, the modem converts **digital signals (0s and 1s)** into **analog signals** so that they can travel through communication media such as telephone lines or cable systems.

Demodulation

When data is received from the internet, the modem converts **analog signals** back into **digital signals** so the computer can understand the information.

Step-by-step process:

1. The computer generates digital data.
2. The modem modulates this digital data into analog signals.
3. Signals travel through communication channels (cable, telephone line, fiber).
4. The receiving modem demodulates the analog signals back into digital form.

5. The computer processes the data.

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3. What is an IP Address?

An **IP Address (Internet Protocol Address)** is a unique numerical identifier assigned to every device connected to a network or the internet.

It helps devices identify and communicate with each other.

Example of an IPv4 address:

192.168.1.1

Types of IP addresses:

- **IPv4** – Uses 32-bit numbers (e.g., 192.168.0.1)
- **IPv6** – Uses 128-bit numbers for a much larger number of devices (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)

IP addresses are essential for routing data to the correct destination on the internet.

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4. What is an ISP?

An **ISP (Internet Service Provider)** is a company that provides internet access to users.

It connects your home or organization to the global internet network.

Examples of ISPs include companies that provide:

- Broadband internet
- Fiber internet
- Mobile data services

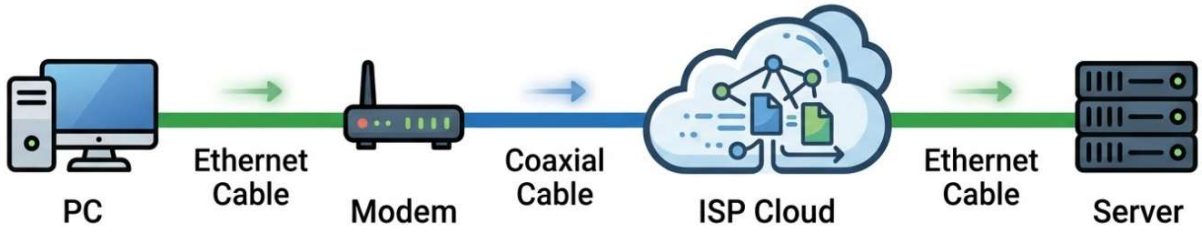
Services provided by ISPs:

- Internet connectivity
- Email services
- Web hosting
- Domain registration
- Network security features

Without an ISP, devices cannot access the internet.

Block Diagram

PC → Ethernet Cable → Modem → Coaxial Cable → ISP Cloud → Ethernet Cable → Server



Hardware Components Required

- **PC-PT:** The user's computer (Digital Source).
- **Cable Modem-PT:** The signal converter (Modulator-Demodulator).
- **Cloud-PT (ISP):** The Service Provider infrastructure.
- **Server-PT:** The remote destination (e.g., a Web Server).

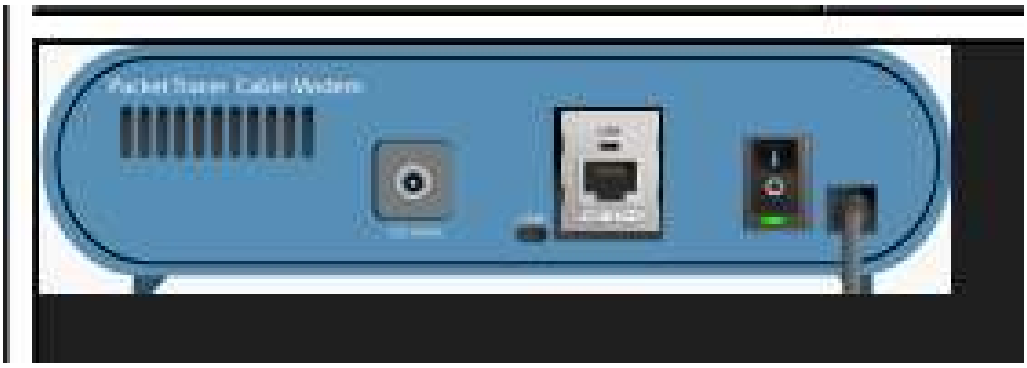
Procedure

Step 1: Physical Topology Design

In this stage, we select the hardware from the device box and place them on the logical workspace.

- **Add End Devices:** Go to the bottom-left "End Devices" category. Drag and drop one PC (PC-PT) and one Server (Server-PT) onto the workspace. Rename the PC by your batch and roll number (e.g. S11_01).
- **Add the ISP Infrastructure:** From the "WAN Emulation" category, drag and drop the Cloud-PT between the PC and the Server. Rename this as "ISP_Cloud".
- **Add the Modem:** Go to "Network Devices" -> "WAN Emulation". Drag and drop the Cable Modem-PT between the PC and the Cloud.

- **Hardware Identification:** Click on the Modem and observe the physical rear view. Note the difference between the small RJ-45 Ethernet port and the threaded Coaxial port.



Step 2: Physical Cabling (Media Selection)

Selecting the correct transmission media is crucial for simulating real-world telecommunication links.

- **PC to Modem:** Select the Copper Straight-Through cable (solid black line). Connect the FastEthernet 0 port of the PC to Port 1 (ethernet port) of the Cable Modem.
- **Modem to ISP:** Select the Coaxial cable (solid blue line). Connect Port 0 (co-axial port) of the Cable Modem to the Coaxial 7 port of the ISP_cloud.
- **ISP to Server:** Select the Copper Straight-Through cable. Connect the Ethernet 6 port of the ISP_Cloud to the FastEthernet 0 port of the Server.

Step 3: ISP Cloud Logic Configuration

The Cloud must be told how to "route" the signal from the analog coaxial side to the digital server side.

- Click on the ISP_Cloud and go to the Config tab.
- In the left-hand menu, look under the INTERFACE heading and click on Ethernet6. Ensure "Cable" is selected under the Provider Network.
- In the left-hand menu, look under the CONNECTIONS heading and click on Cable.
- In the drop-down menus, select Coaxial 7 for the first field and Ethernet 6 for the second field.
- Click the Add button. You should now see Coaxial7 <==> Ethernet6 appearing in the mapping table.

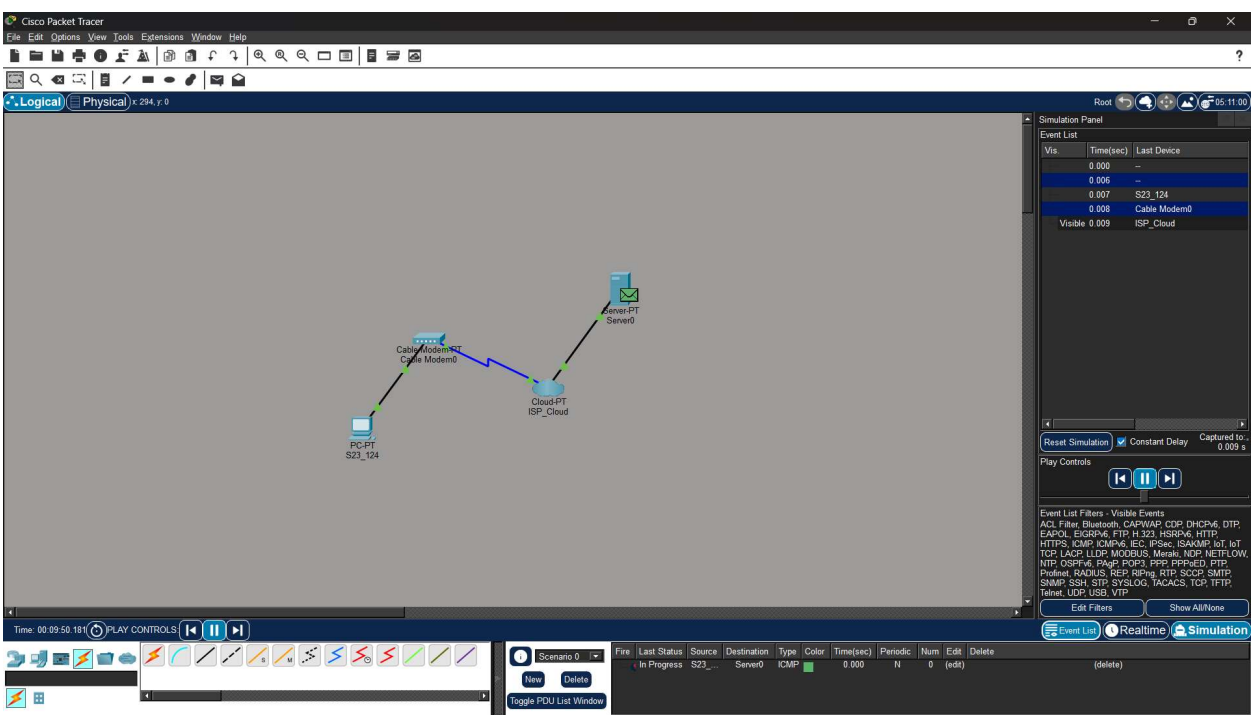
Step 4: IP Addressing (Logical Setup)

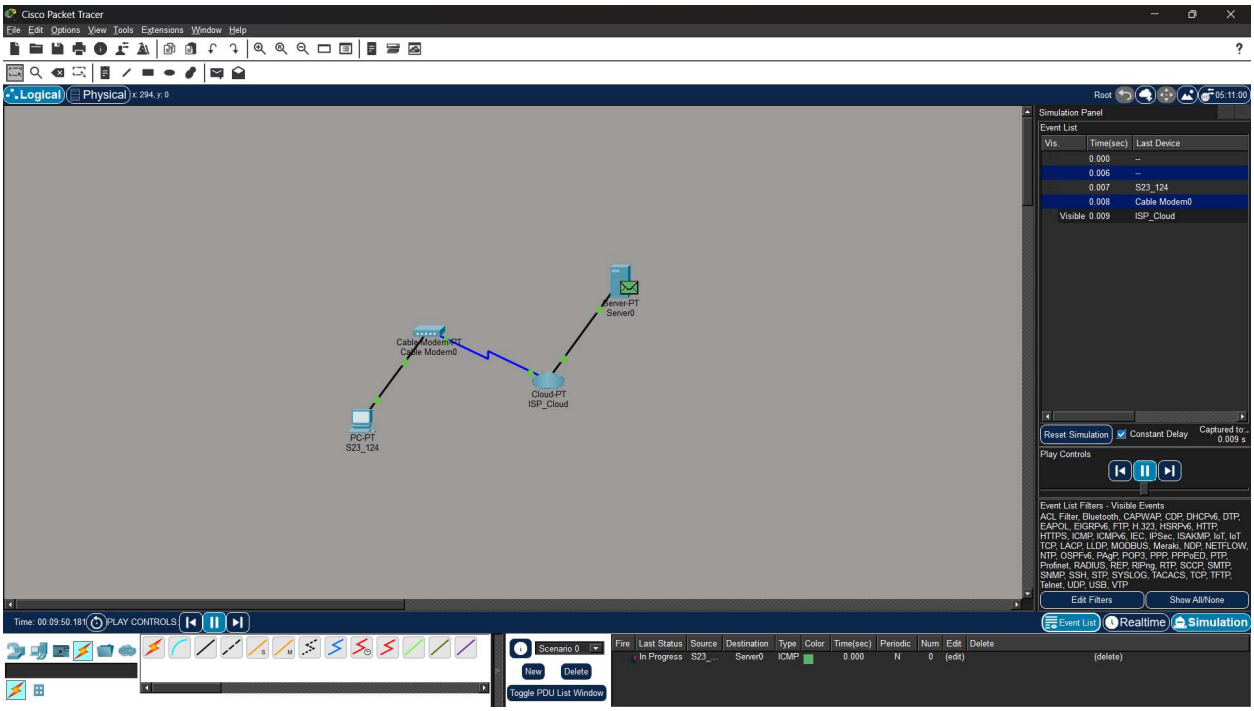
Both the source (PC) and destination (Server) must belong to the same logical network to communicate without a router.

- **Configure PC:** Click PC -> Desktop -> IP Configuration. Set the Static IP address to 192.168.1.1 and Subnet Mask to 255.255.255.0.
- **Configure Server:** Click Server -> Desktop -> IP Configuration. Set the IP address to 192.168.1.2 and Subnet Mask to 255.255.255.0.

Step 5: Testing and Packet Observation

- **Real-time Testing:**
  - Open the Command Prompt on the PC. Type ping 192.168.1.2. If you see "Reply from...", the modem is correctly modulating and demodulating the signal.
- **Simulation Mode:** Click the Simulation tab (bottom-right).
  - Click Edit Filters and select only ICMP.
  - Click the Add Simple PDU (Envelope icon), click on the PC, then click on the Server.
  - Click the Play/Forward button to observe the packet travel from the digital Ethernet link to the analog Coaxial link.





Observations

- **Medium Change:** It is observed that the data travels as a digital signal until it reaches the Modem.
- **Signal Conversion:** Upon hitting the Modem, the data is converted to travel over the **Coaxial medium** (simulating analog transmission).
- **Successful Loop:** The Server successfully receives the modulated signal, ISP cloud demaps the return packet back to the specific Coaxial line, allowing the PC to receive a successful reply.

Conclusion

The experiment successfully demonstrates that a Modem is an essential interface for internet connectivity. It proves that digital end-devices cannot communicate directly over analog WAN infrastructure without the modulation/demodulation hardware.